System Design Document

Project Title:

AI Study Material Generator & Assistant

Course Title:

Software Development III

Course Code:

CSE 4104

Section:

7C

Team Name:

CSE4104-7C-T03

Submitted By	***Submitted To:***
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1. Project Overview

Project Description

AI Study Material Generator is an intelligent educational platform that helps students convert study materials into useful learning resources automatically.

The system allows users to upload study documents and generate:

- Summaries
- Multiple Choice Questions (MCQs)
- Flashcards
- Quizzes
- AI Chat Assistance

The project uses Artificial Intelligence through OpenAI API to analyze educational content and generate personalized study materials.

Objectives

• Generate study materials automatically

• Reduce manual note preparation time

• Improve revision efficiency

• Provide AI-powered educational assistance

• Enhance student productivity

2. System Architecture Diagram

Architecture Components

Frontend Layer

- React.js
- Tailwind CSS
- JavaScript

Backend Layer

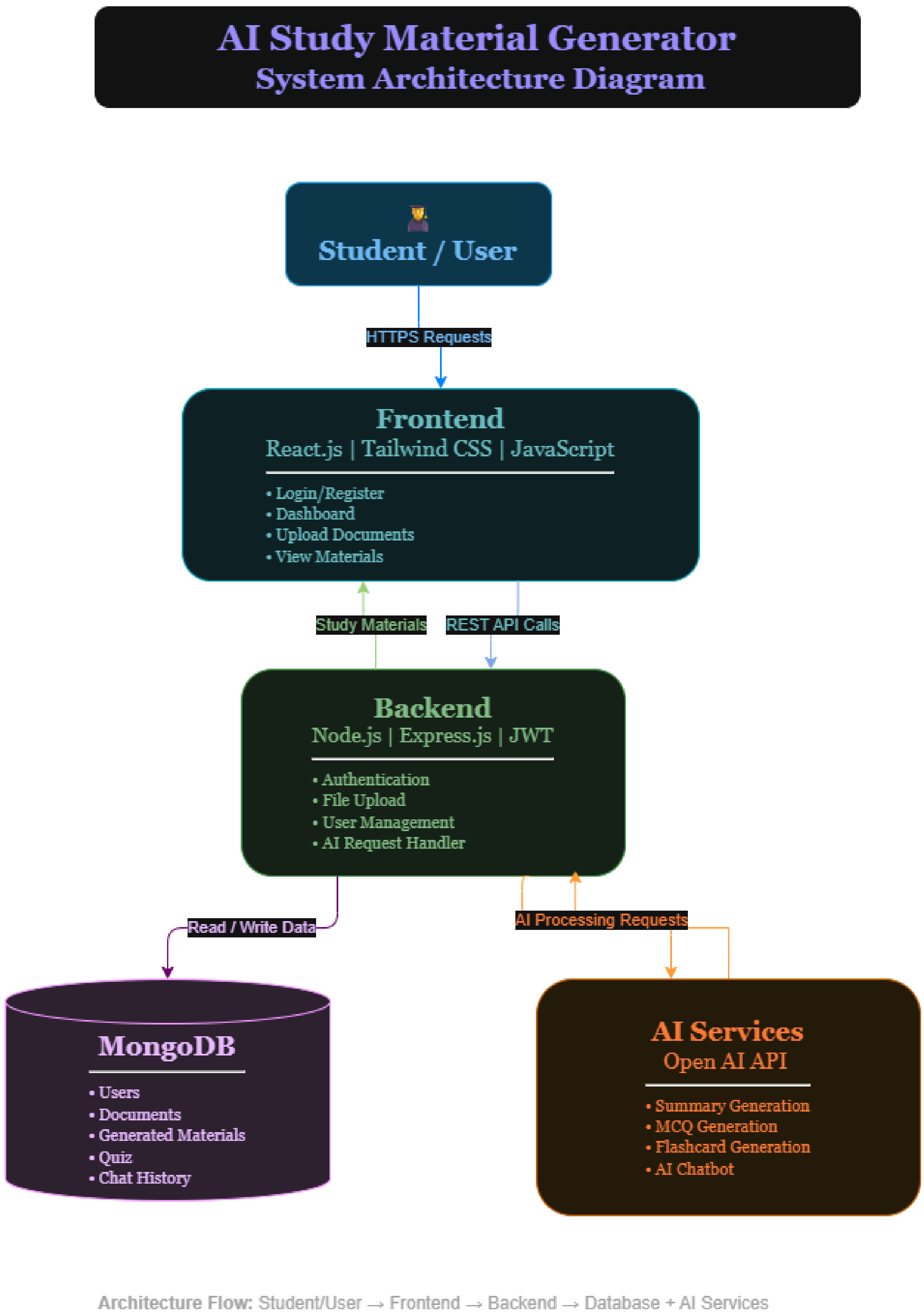
- Node.js
- Express.js
- JWT Authentication

Database Layer

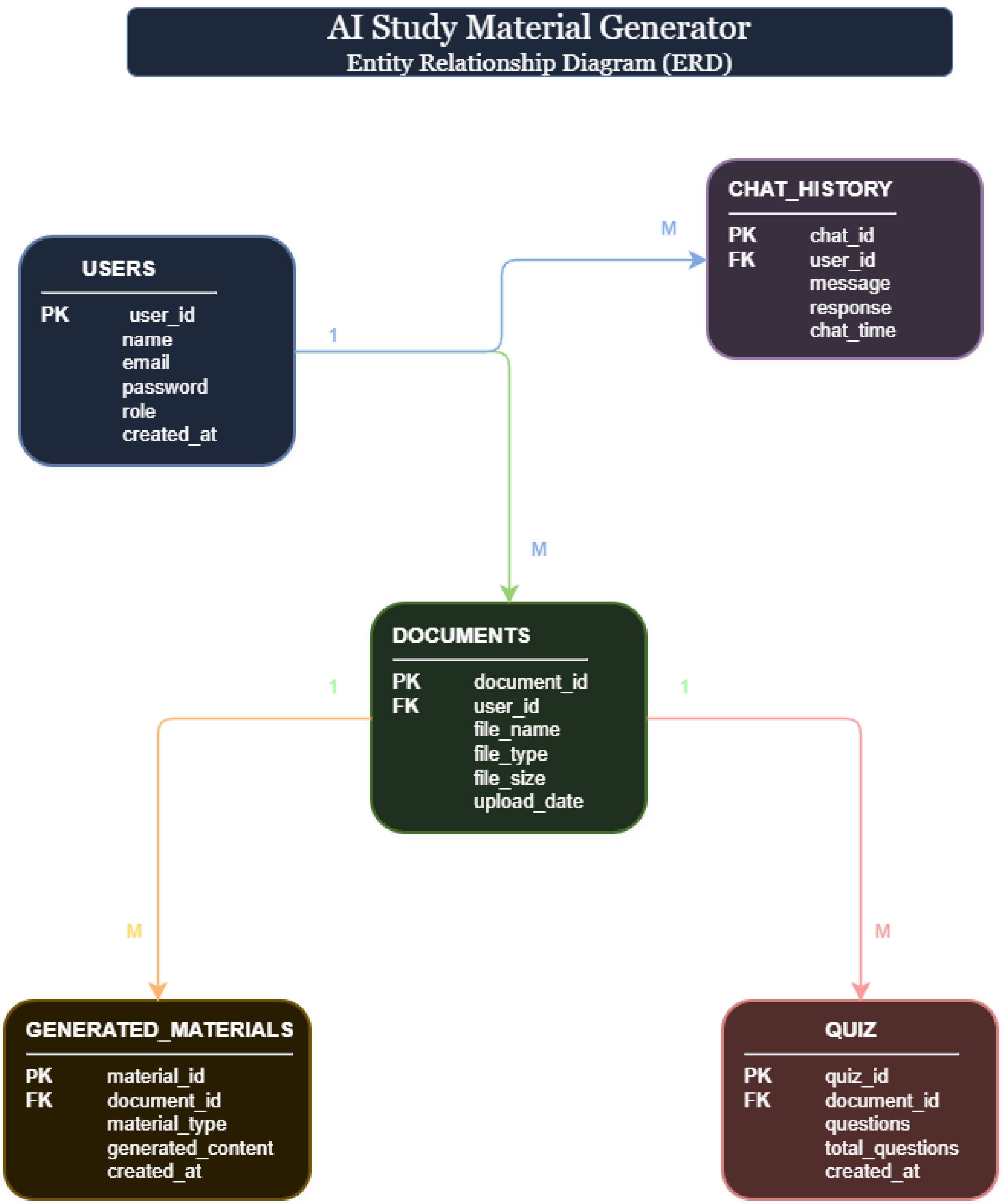
- MongoDB

AI Services

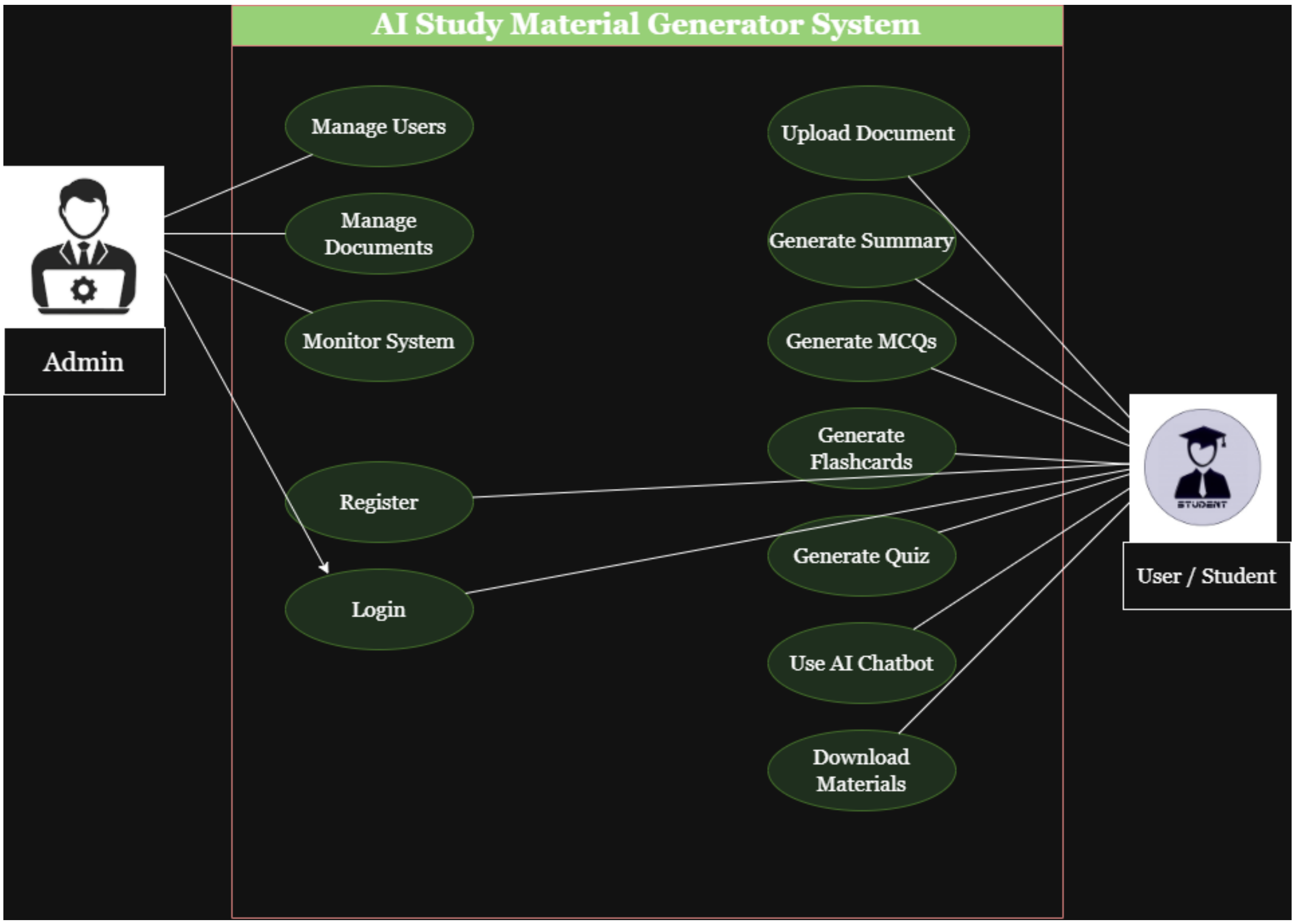
- OpenAI API


AI Study Material Generator
System Architecture Diagram
Student / User
HTTPS Requests
Frontend
React.js | Tailwind CSS | JavaScript
• Login/Register
• Dashboard
• Upload Documents
• View Materials
Study Materials
REST API Calls
Backend
Node.js | Express.js | JWT
• Authentication
• File Upload
• User Management
• AI Request Handler
Read / Write Data
AI Processing Requests
MongoDB
• Users
• Documents
• Generated Materials
• Quiz
• Chat History
AI Services
Open AI API
• Summary Generation
• MCQ Generation
• Flashcard Generation
• AI Chatbot
Architecture Flow: Student/User → Frontend → Backend → Database + AI Services

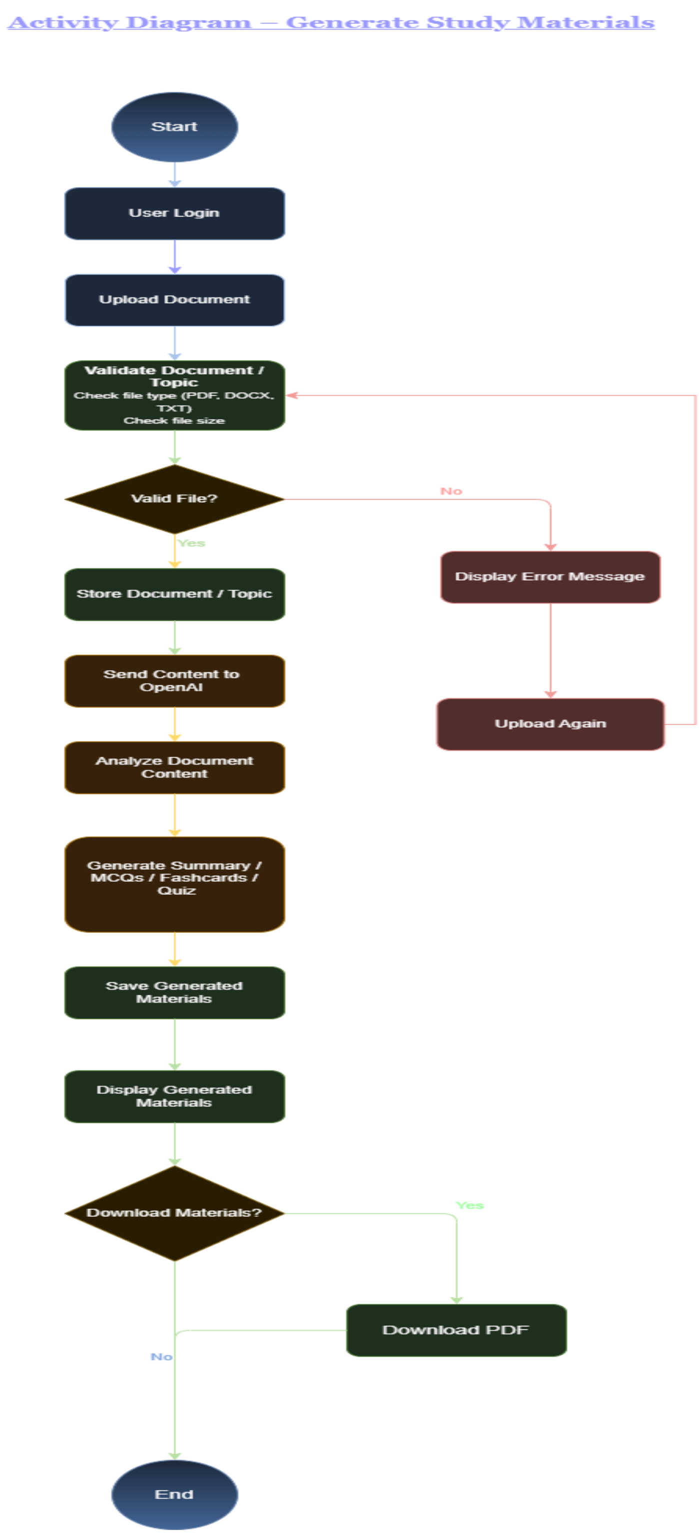
3. Updated ER Diagram (Database Design)



4. Updated Use Case Diagram



5. Activity Diagram



6. API Design Document

Authentication APIs

POST /api/auth/register

Purpose:
Register a new user.

Input:

```
{
"name": "string",
"email": "string",
"password": "string"
}
```

Output:

```
{
"message": "Registration Successful"
}
```

POST /api/auth/login

Purpose:
Authenticate user.

Input:

```
{
"email": "string",
"password": "string"
}
```

Output:

```
{
"token": "jwt_token"
}
```

User APIs

GET /api/user/profile

Purpose:
Retrieve user profile.

PUT /api/user/profile

Purpose:
Update profile information.

Document APIs

POST /api/documents/upload

Purpose:
Upload study material.

Input:
PDF/DOCX/TXT file

Output:

{
"document_id": 101
}

GET /api/documents

Purpose:
Retrieve uploaded documents.

AI APIs

POST /api/ai/summary

Purpose:
Generate document summary.

Input:

{
"document_id": 101
}

Output:

{
"summary": "Generated Summary"
}

POST /api/ai/mcq

Purpose:
Generate MCQs.

Output:

{
"mcqs": []
}

POST /api/ai/flashcards

Purpose:
Generate Flashcards.

Output:

{
"flashcards": []
}

POST /api/ai/chat

Purpose:
Interact with AI chatbot.

Input:

{
"message": "Explain Machine Learning"
}

Output:

{
"response": "AI generated answer"
}

7. AI Integration Workflow

Why AI is needed:

Manually creating summaries, MCQs, flashcards, and quizzes from lecture notes is time-consuming. AI using NLP can read raw document text and automatically extract key concepts, rephrase content into summaries, and generate relevant questions, drastically reducing student preparation time and providing personalized assistance.

AI Service/Model Used:

OpenAI API (e.g., GPT-4 / GPT-4o-mini via Chat Completions API) — chosen for strong text comprehension, summarization, and structured Q&A generation capabilities.

Input to the AI System:

- Extracted text content from uploaded PDF/DOCX/TXT (chunked if large)
- A task-specific prompt (e.g., "Summarize this text into bullet points," "Generate 10 MCQs with 4 options and correct answer")
- For chatbot: user's question + relevant document context

Output from the AI System:

- Structured JSON or text response: summaries, MCQ arrays (question/options/answer), flashcard pairs, quiz sets, or chatbot replies

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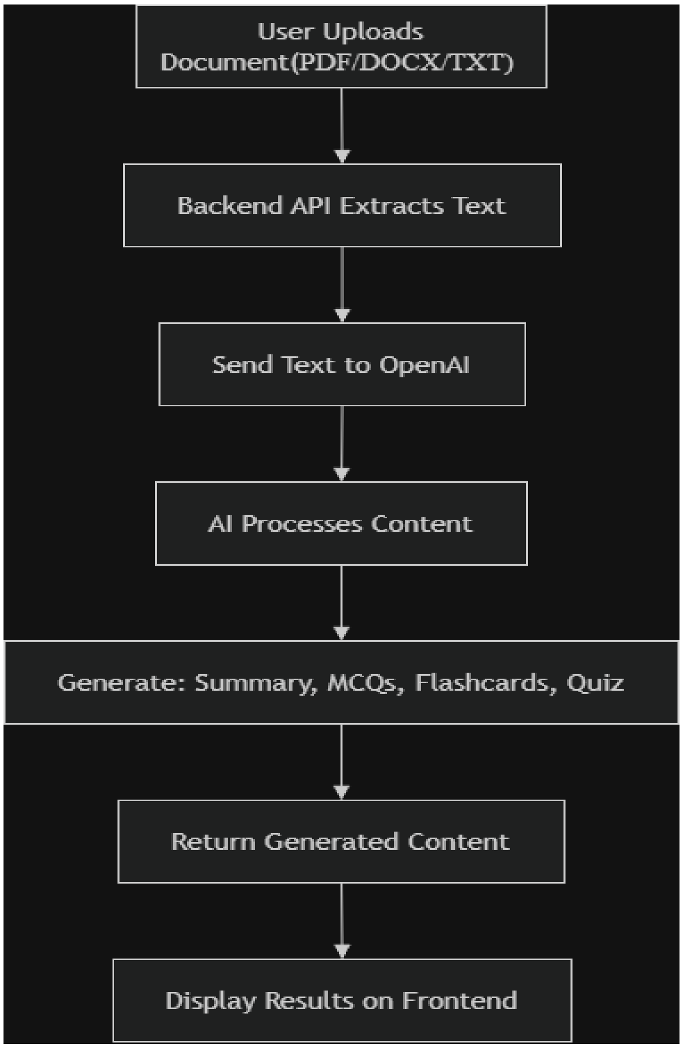
AI Workflow



Benefits of AI Integration

- Saves study time
- Creates personalized learning materials
- Improves revision efficiency
- Enhances student productivity
- Provides instant educational assistance

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9. References

1. OpenAI Documentation
2. Google Gemini API Documentation
3. MongoDB Documentation
4. React.js Documentation
5. Node.js Documentation
6. Express.js Documentation